

# VLSI Design & Physical Verification Portfolio

Analog Signal Processing, Custom Standard Cells, and  
Memory Architecture

EDA Toolchain: Cadence Virtuoso Suite & LTspice

*Compiled Document of Post-Layout Extraction and Hardware Characterization*

## **Core Competencies Demonstrated:**

Continuous-Time Analog Signal Processing  
Operational Mirrored Amplifiers & Translinear Circuits  
Schematic Entry, Logic Verification & Custom Layout  
DRC, LVS, and RC Parasitic Extraction  
Logical Effort & Pipeline Delay Analysis  
Memory Array Architecture (ROM)

2025 – 2026

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# Phase 1

## Analog IC Design & Current-Mode Signal Processing

This section details the design and characterization on both **LTspice** and **Hardware** of continuous-time analog signal processing blocks, specifically focusing on translinear principles, Operational Mirrored Amplifiers (OMA), and Current Conveyor architectures (CFOA/CCII+).

### 1.1 Operational Mirrored Amplifier (OMA) & Current Followers

#### 1.1.1 Design Objective

To realize and characterize current followers using Operational Mirrored Amplifiers, specifically comparing the performance of a Simple Current Mirror against a Cascode Current Mirror to evaluate Early effect mitigation and bandwidth extension[cite: 141].

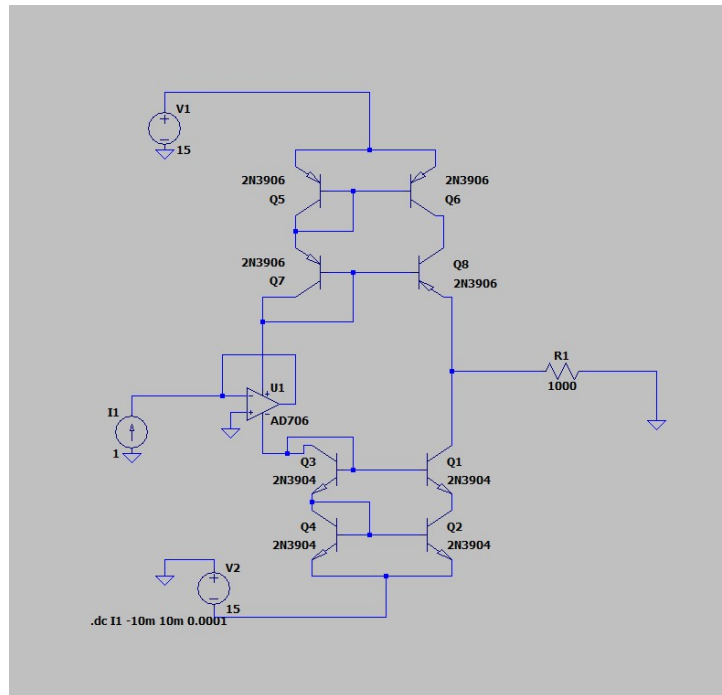


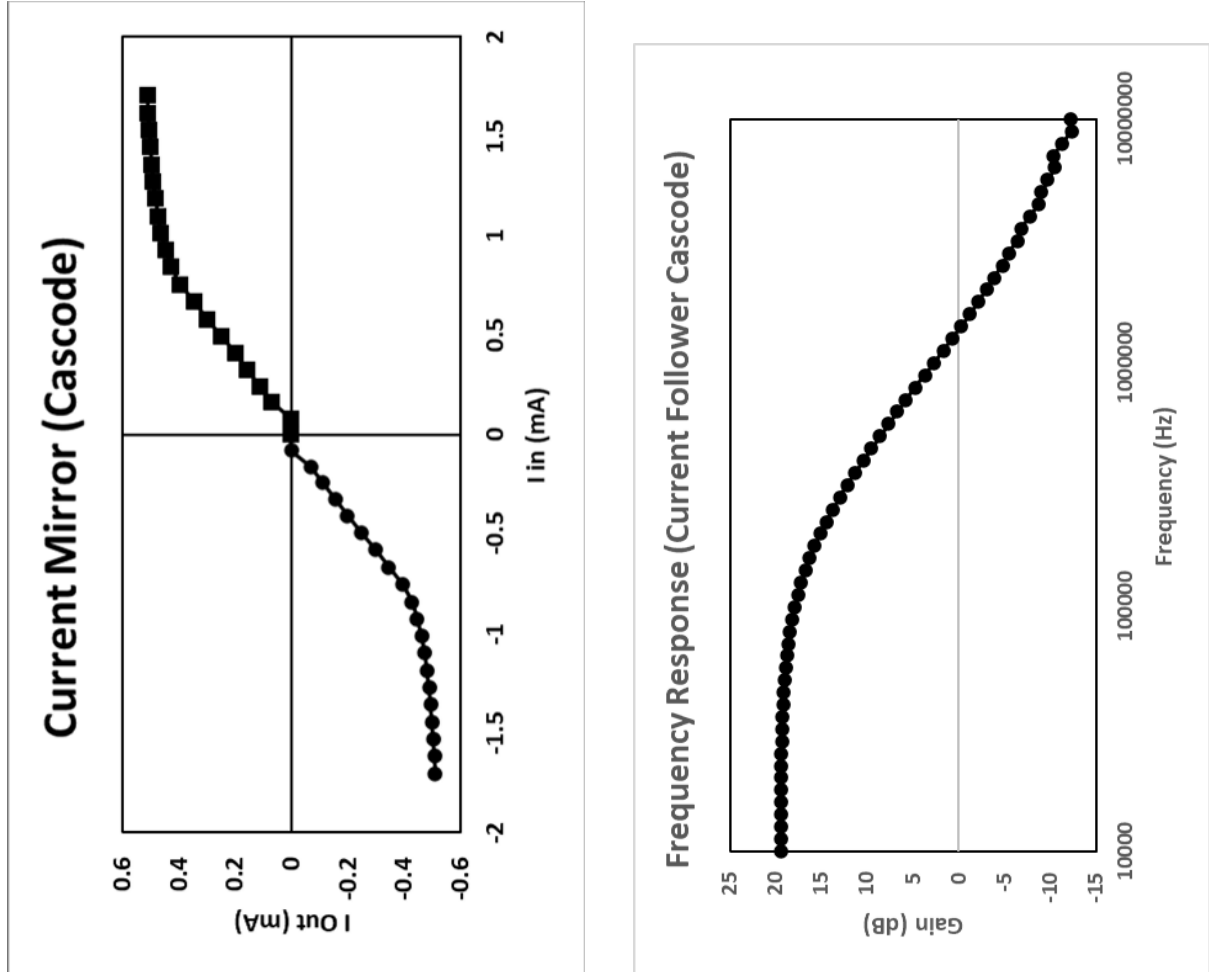
Figure 1.1: Full Schematic of the Cascode Operational Mirrored Amplifier (OMA)

#### 1.1.2 Architecture & Implementation

The implementation utilized 2N3904 (NPN) and 2N3906 (PNP) BJT arrays[cite: 207].

- **Simple Current Mirror:** While  $I_{out} \approx I_{in}$ , this topology suffers from the Early effect (channel length modulation in MOS equivalents), where  $I_{out}$  erroneously increases as the output voltage  $V_{CE}$  increases[cite: 207].
- **Cascode Current Mirror:** This topology utilizes four transistors[cite: 207]. The bottom pair mirrors the current, while the top pair acts as a shield, isolating the mirroring devices from load voltage variations[cite: 207].

### 1.1.3 Performance Characterization



(a) DC Transfer Characteristic ( $I_{out}$  vs  $I_{in}$ )

(b) AC Frequency Response

Figure 1.2: Performance Characteristics of the Cascode Current Follower

#### Observations:

- **Impedance & Linearity:** The Cascode configuration successfully shielded the current-setting transistors, drastically increasing the output impedance and maintaining a highly linear  $I_{out}$  vs  $I_{in}$  relationship independent of load variations[cite: 250].
- **Frequency Response:** AC analysis confirmed that the Cascode configuration successfully mitigated the Miller effect, resulting in a significantly extended operational bandwidth compared to the Simple Current Mirror[cite: 207, 250].

## 1.2 Constant Bandwidth Variable Gain Amplifier (CFOA)

### 1.2.1 Design Objective

To demonstrate the decoupling of gain and bandwidth using an Operational Mirrored Amplifier / Current Feedback Op-Amp (CFOA), directly contrasting the inherent Gain-Bandwidth Product (GBW) limitations of standard Voltage Feedback Amplifiers (VFA)[cite: 279, 291].

### 1.2.2 Theory & Validation

In a standard VFA, the bandwidth drops proportionally as closed-loop gain increases[cite: 288, 290]. However, in a CFOA, the bandwidth is determined almost entirely by the feedback resistor ( $R_f$ )[cite: 295]. By locking  $R_f$  as a constant and exclusively varying the ground-referencing resistor ( $R_g$ ), the voltage gain ( $A_v = 1 + R_f/R_g$ ) can be modulated without sacrificing bandwidth[cite: 298, 300, 305].

Table 1.1: Hardware Validation of CFOA Constant Bandwidth

| Gain Configuration | Peak Gain (dB) | Measured Hardware B.W.          |
|--------------------|----------------|---------------------------------|
| $A_v = 10$         | 17 dB          | 200 kHz* [cite: 267]            |
| $A_v = 20$         | 22.5 dB        | 200 kHz* [cite: 267]            |
| $A_v = 30$         | 31.6 dB        | Constant Range [cite: 444, 446] |

*\*Note: While LTspice simulations yielded a stable bandwidth of  $\approx 891$  kHz, hardware prototype bandwidths were constrained to a lower constant range due to breadboard parasitic strays and unmatched discrete device capacitances[cite: 444, 446]. The theoretical proof of concept, however, remained valid across all gain steps.*

## 1.3 Translinear Current Squarer

### 1.3.1 Design Objective

To design, simulate, and experimentally validate a Current Squarer circuit utilizing the translinear principle to achieve a precise parabolic input-output transfer characteristic[cite: 523, 528].

### 1.3.2 Architecture & Implementation

The circuit leverages the logarithmic-exponential relationship of bipolar junctions in a translinear loop to perform analog computation. To ensure tight thermal coupling and  $V_{BE}$  matching, the hardware implementation utilized monolithic CA3086 and CA3096 BJT arrays[cite: 526, 584].

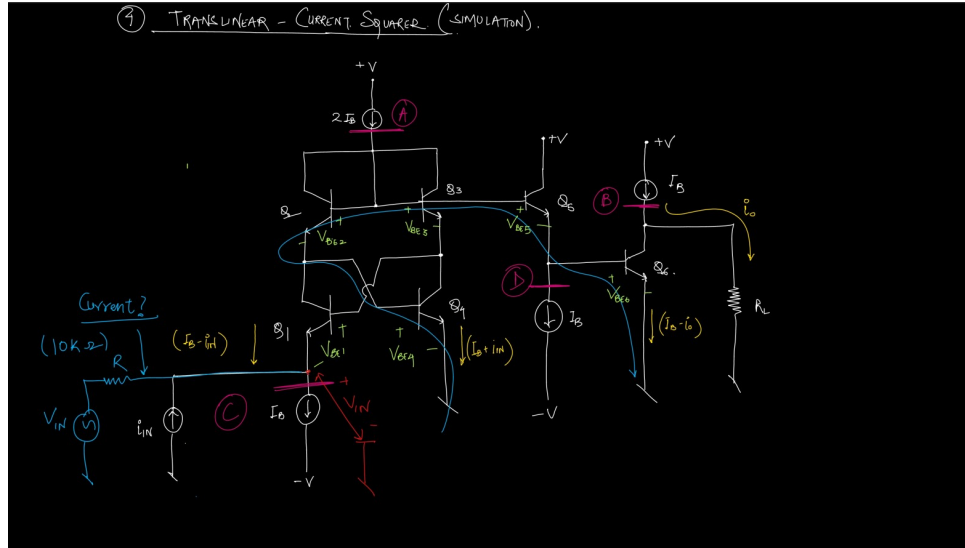
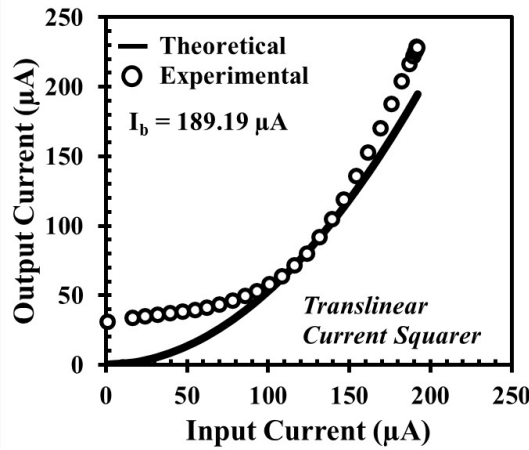
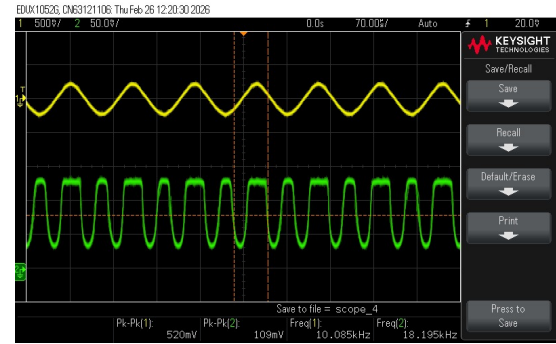


Figure 1.3: Translinear Current Squarer Schematic and Nodal Analysis



(a) DC Transfer Characteristic



(b) Transient Frequency Doubling

Figure 1.4: Experimental Validation of the Translinear Squarer

### Observations:

- **DC Characterization:** The experimental DC sweep closely tracks the theoretical square-law curve ( $I_{out} \propto I_{in}^2$ ) for a measured bias current of 189.19  $\mu\text{A}$ .
- **Transient Limitations:** While current mirroring operated successfully, high-frequency signal doubling in the physical hardware exhibited degradation due to non-ideal junction diffusion differences and stray array parasitics [cite: 661, 663, 671].

## 1.4 Negative Impedance Converter (NIC) using CCII+

### 1.4.1 Design Objective

To realize and verify a Negative Impedance Converter (NIC) utilizing a Second-Generation Current Conveyor (CCII+) based on the AD844 architecture,

demonstrating its application as a fundamental building block for Voltage Controlled Resistors (VCR).

### 1.4.2 Theory & Schematic

The design utilizes a CCII+, characterized by its ideal terminal matrix relations:  $V_x = V_y$ ,  $I_z = I_x$ , and  $I_y = 0$ .

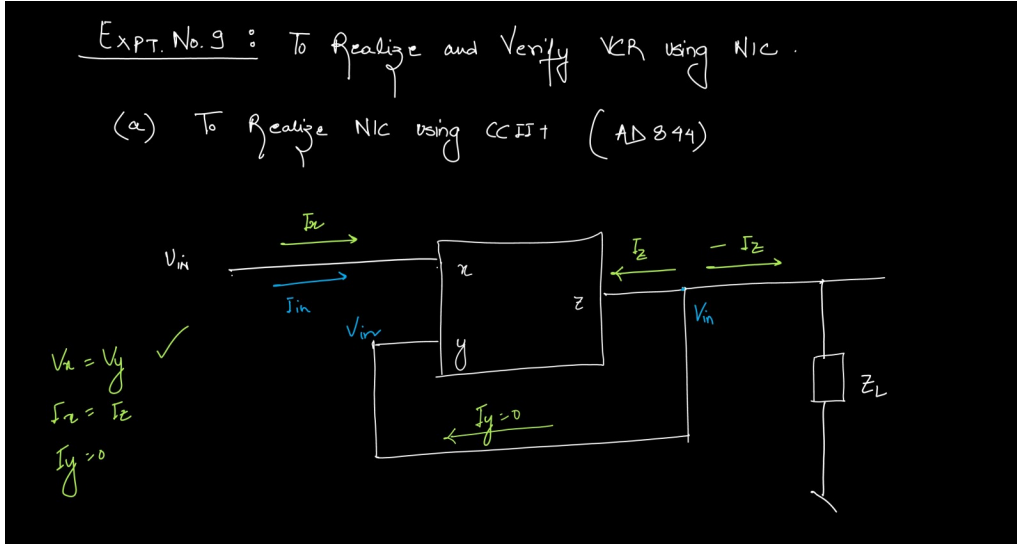


Figure 1.5: CCII+ based Negative Impedance Converter (NIC) Schematic

By utilizing the current tracking between the X and Z terminals alongside the voltage tracking between the Y and X terminals, the feedback topology effectively reflects the load impedance  $Z_L$  with a negative sign at the input port, yielding  $Z_{in} = -Z_L$ . This continuous-time property is highly advantageous for parasitic resistance cancellation and high-Q oscillator design.

## 1.5 Two-CFOA Universal Multifunction Filter

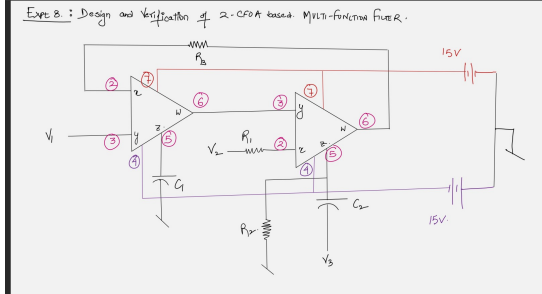
### 1.5.1 Design Objective

To design a voltage-mode universal biquad filter using two AD844 Current Feedback Operational Amplifiers capable of realizing Low-Pass (LPF), High-Pass (HPF), Band-Pass (BPF), Notch, and All-Pass (APF) responses from a single unified topology.

### 1.5.2 Architecture & Transfer Function

Unlike standard Voltage Feedback Op-Amps (VFA), the CFOA avoids severe Gain-Bandwidth (GBW) product limitations. The universal topology accepts three input voltages to synthesize the desired output.





(a) Two-CFOA Circuit Topology

Images/cfoa filter2.jpg

(b) Transfer Function &amp; Tuning Parameters

Figure 1.6: Theoretical Nodal Analysis and Mathematical Derivation

The generalized transfer function derived for this architecture is:

$$V_o = \frac{\frac{1}{C_1 C_2 R_1 R_3} V_1 - \frac{s}{C_2 R_1} V_2 + s^2 V_3}{s^2 + \frac{1}{C_2 R_2} s + \frac{1}{C_1 C_2 R_1 R_3}}$$

### 1.5.3 Filter Specifications & Tuning

The filter was mathematically tuned for a center frequency of 15.9 kHz and a Butterworth quality factor of  $Q = 1/\sqrt{2}$ .

- **Passive Components:**  $C_1 = C_2 = 1$  nF,  $R_1 = R_3 = 10$  k $\Omega$ ,  $R_2 = 7.07$  k $\Omega$ .
- **Mode Selection Routing:**
  - **LPF:**  $V_2 = 0$ ,  $V_3 = 0$ ,  $V_1 = V_{in}$
  - **HPF:**  $V_1 = 0$ ,  $V_2 = 0$ ,  $V_3 = V_{in}$
  - **BPF:**  $V_1 = 0$ ,  $V_3 = 0$ ,  $V_2 = V_{in}$
  - **Notch:**  $V_2 = 0$ ,  $V_1 = V_{in}$ ,  $V_3 = V_{in}$

### 1.5.4 Simulation & Verification

AC sweep simulations were conducted using SPICE models for the AD844AN to validate the derived transfer function.

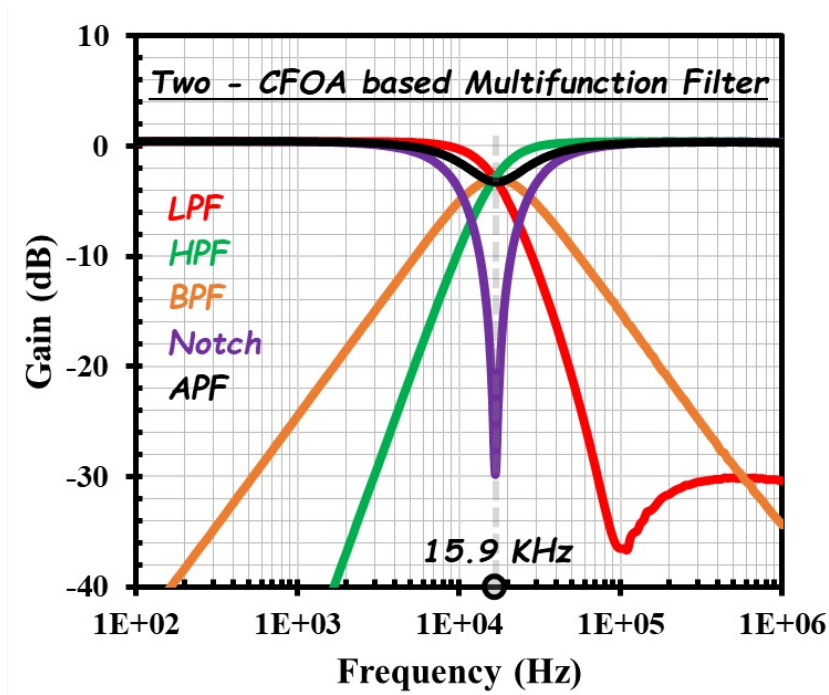


Figure 1.7: AC Response of the Multifunction Filter showing distinct operating modes

**Observations:** The frequency response precisely matches the theoretical hand-calculations. The Bode plots verify that the center frequency intersects exactly at the target 15.9 kHz across all operating modes. This demonstrates excellent stability and high-Q performance without the bandwidth degradation typical of standard op-amp implementations.

# Phase 2

## Standard Cell Physical Design & Characterization

This section details the complete physical design flow—from schematic entry to post-layout simulation with RC extraction—for fundamental CMOS standard cells.

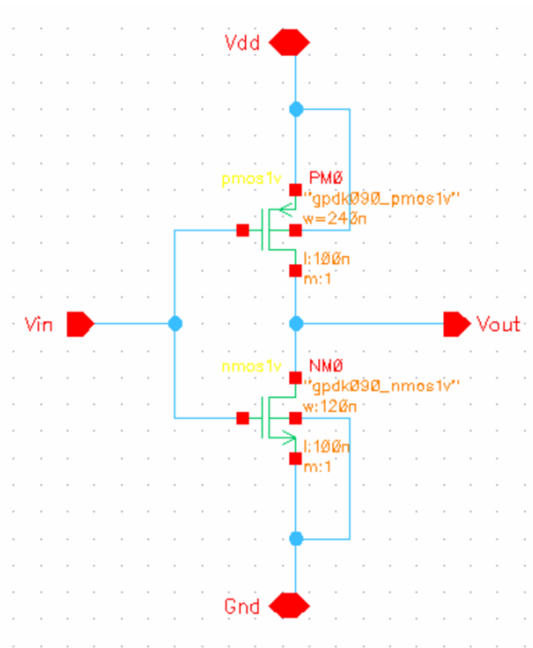
### 2.1 CMOS Inverter Implementation

#### 2.1.1 Design Objective

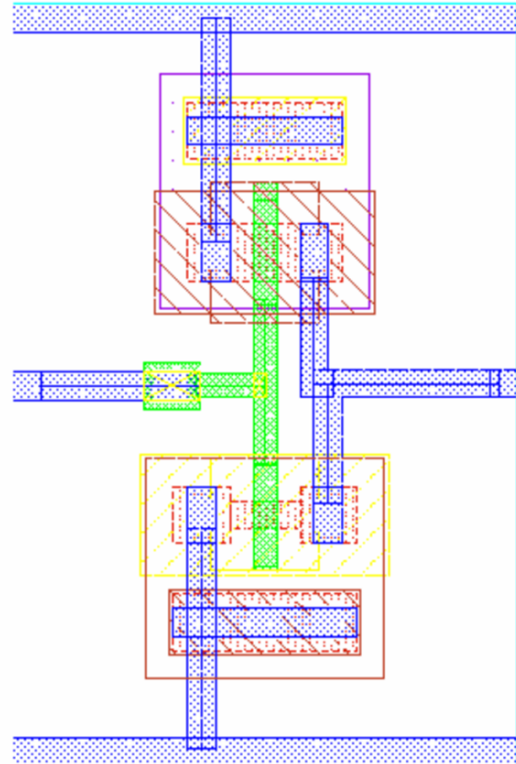
To design the schematic and physical layout of a baseline CMOS Inverter, analyze the Voltage Transfer Characteristics (VTC), and evaluate propagation delays pre- and post-layout.

#### 2.1.2 Architecture & Implementation

The layout structure places the PMOS in an N-Well and the NMOS directly in the P-substrate. Poly-silicon is routed for the gates, with Metal 1 utilized for local interconnects ( $V_{DD}$ , GND, Input, Output).



(a) Schematic



(b) Physical Layout

Figure 2.1: CMOS Inverter Design Views

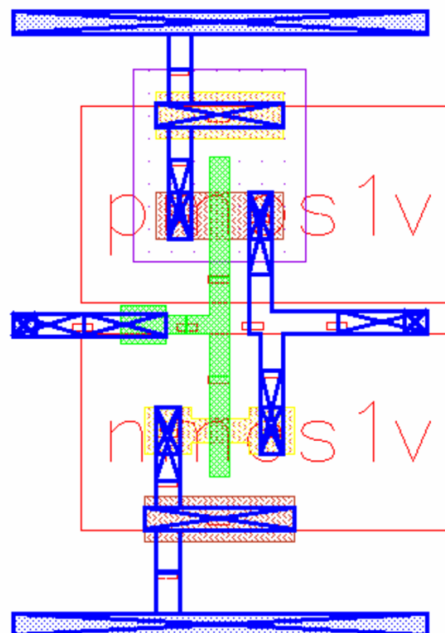


Figure 2.2: LVS Output Log: Confirming Layout Matches Schematic

## 2.1.3 Post-Layout Delay Analysis

RC extraction was performed to model the finite resistance and capacitance of interconnects. Post-layout simulations highlight the impact of these parasitics on signal integrity.

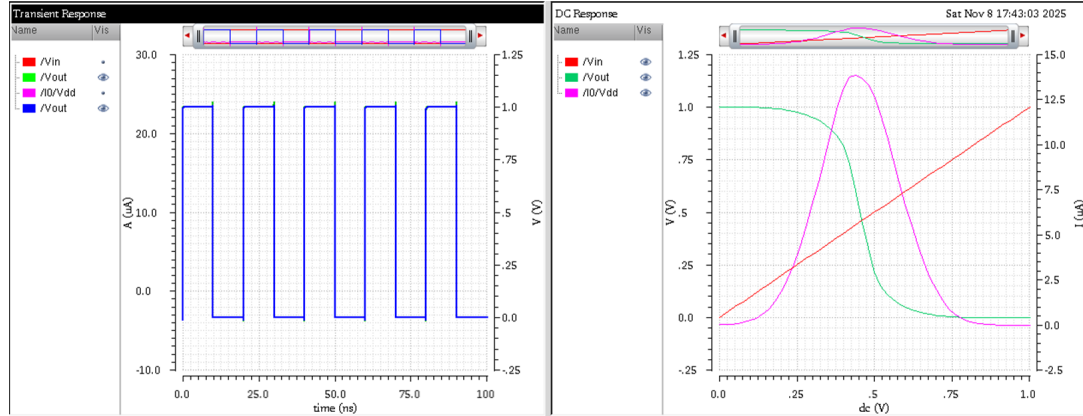


Figure 2.3: Post Layout Transient Response

**Observations:** The switching threshold ( $V_M$ ) was measured at approximately  $V_{DD}/2$ , indicating balanced transistor sizing. Delays calculated via the ADE calculator confirmed the layout's speed performance, with parasitic elements causing expected minor degradation compared to ideal schematic simulations.

## 2.2 NAND2 Gate Implementation

### 2.2.1 Design Objective

To construct a 2-input CMOS NAND gate, perform layout generation, ensure DRC/LVS compliance, and extract RC parasitics to verify the High-to-Low ( $t_{pHL}$ ) discharge path.

### 2.2.2 Physical Layout & LVS

The Pull-Down Network (PDN) utilizes series NMOS devices, which dictate the  $t_{pHL}$  performance.

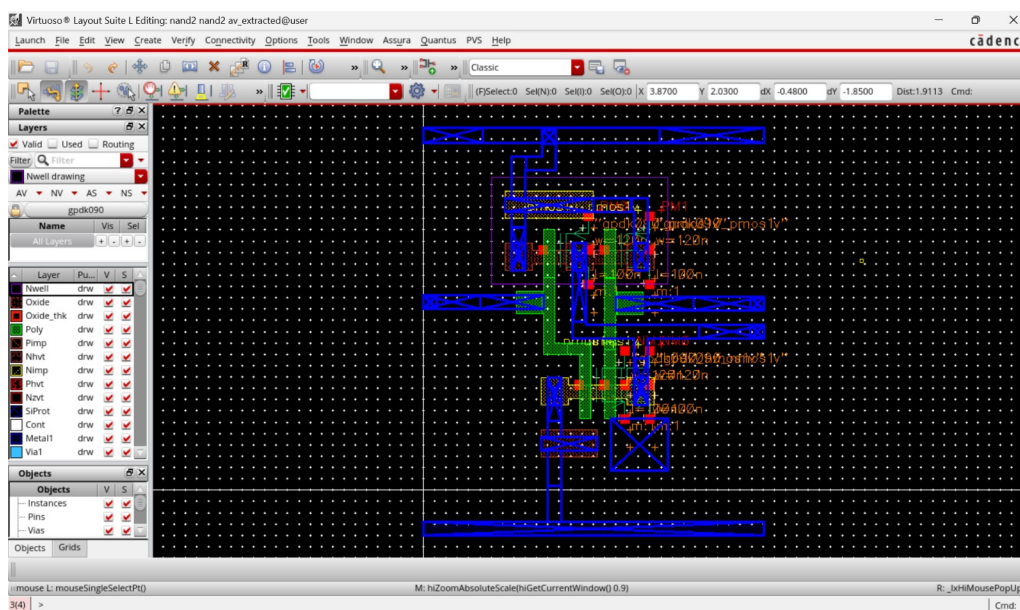


Figure 2.4: NAND2 Gate Custom Layout

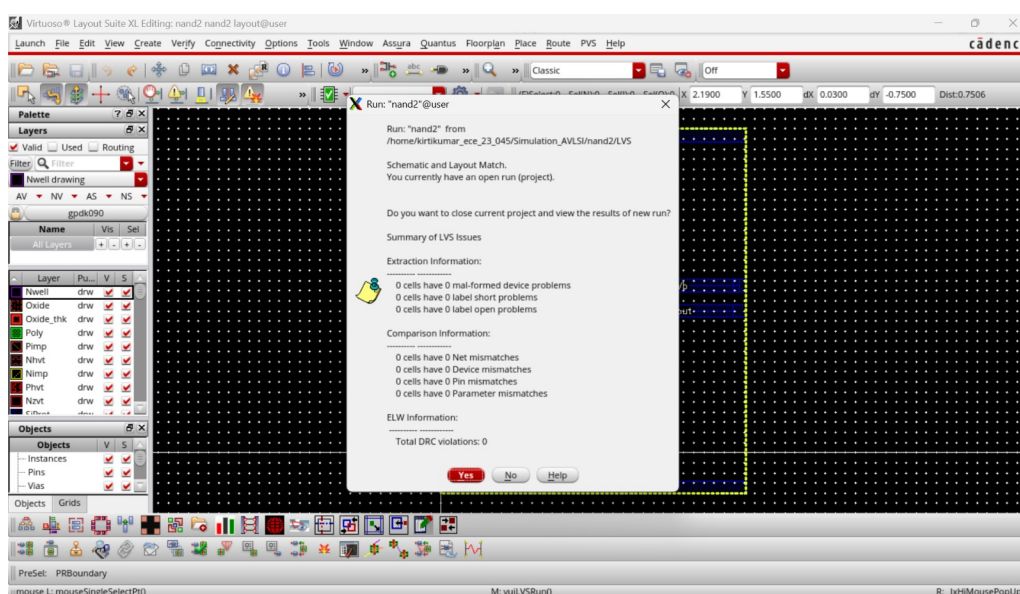


Figure 2.5: NAND2 LVS and RC Extraction Clearance

### 2.2.3 Verification & Signal Analysis

The transient response illustrates the superposition of schematic output (/vout\_Schematic) versus the post-layout extracted output (/vout\_Layout).

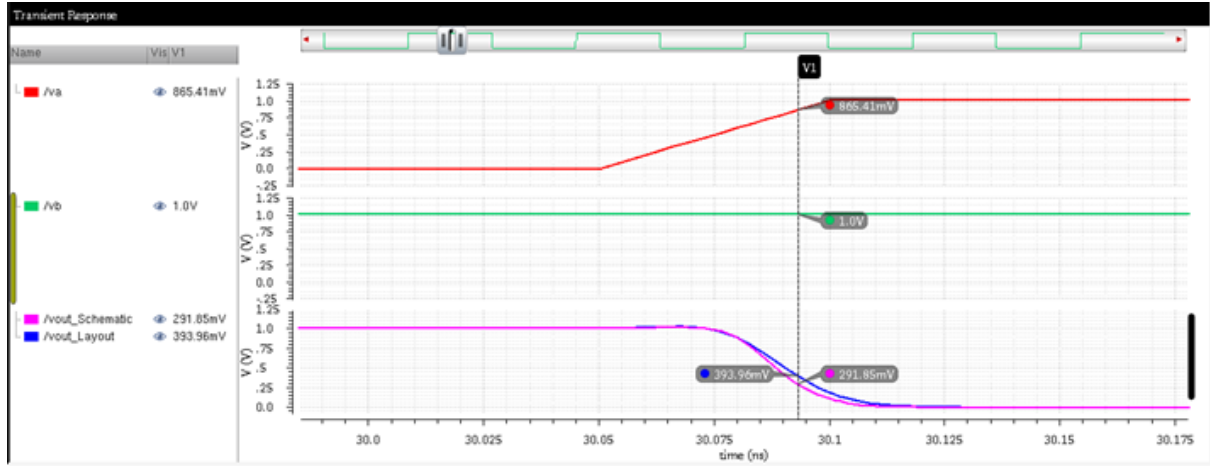


Figure 2.6: NAND2 Pre vs. Post-Layout Simulation

**Observations:** The layout output lags slightly behind the schematic output. At a measured instant, the schematic design discharged to 291.85 mV, while the post-layout output reached 393.96 mV. This shift isolates the impact of the parasitic capacitance ( $C_{par}$ ) in the diffusion regions and metal routing.

## 2.3 NOR2 Gate Implementation

### 2.3.1 Design Objective

To implement a 2-input CMOS NOR gate, characterizing the series PMOS Pull-Up Network (PUN) and comparing its parasitic loading to the NAND2 equivalent.

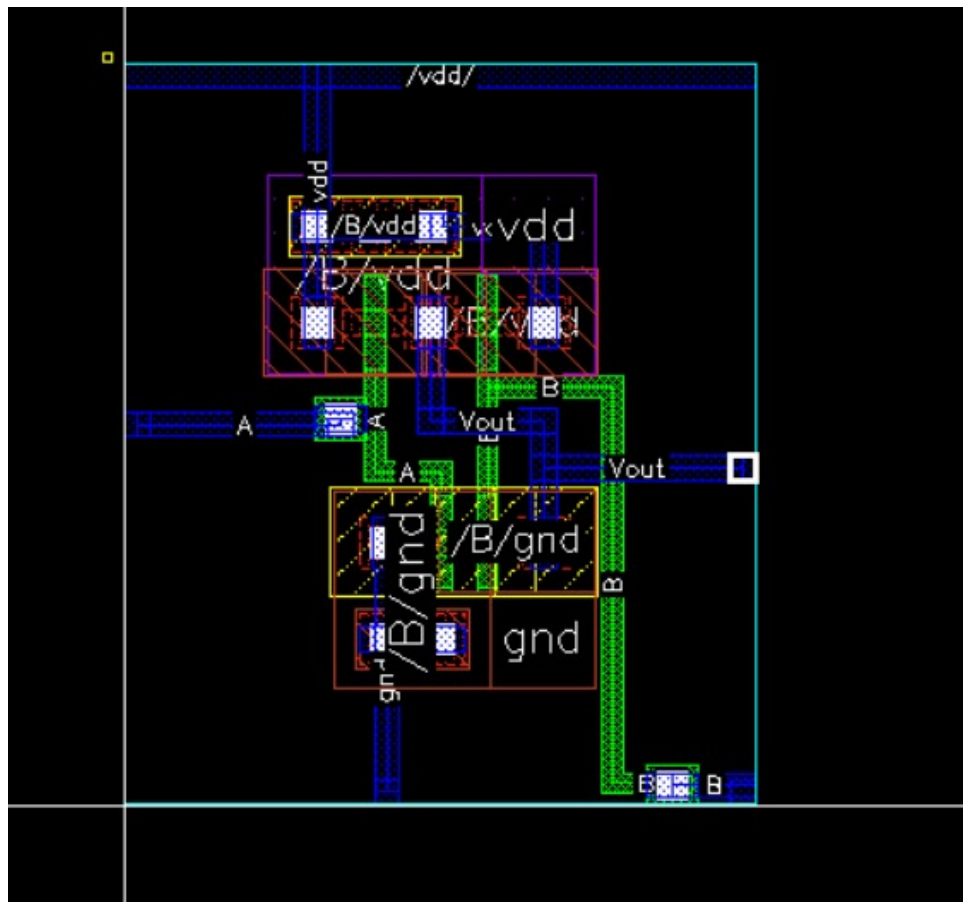


Figure 2.7: NOR2 Gate Physical Layout

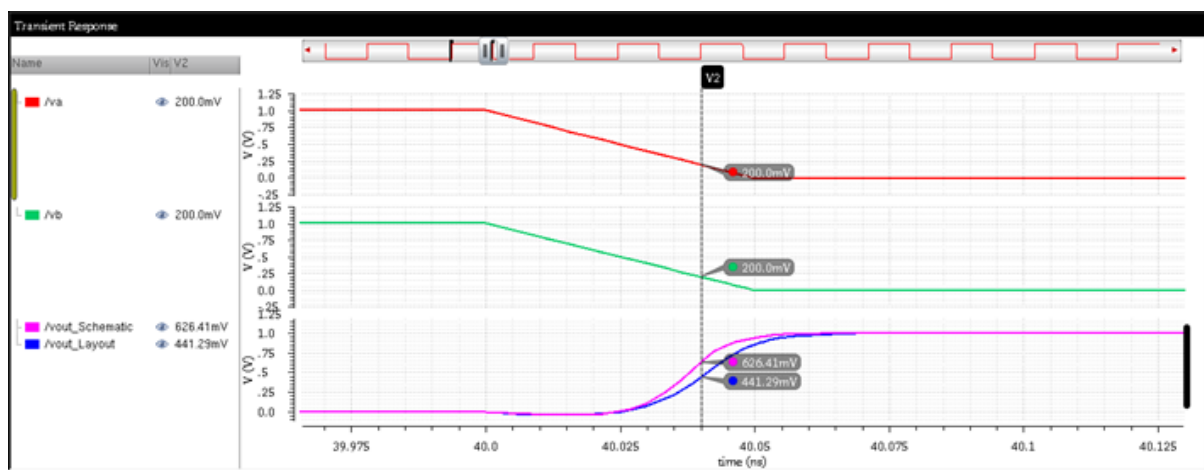


Figure 2.8: NOR2 Pre vs. Post-Layout Simulation

**Observations:** The post-layout simulation demonstrates a finite propagation delay ( $t_{pd}$ ) slightly larger than the schematic due to extracted parasitic capacitance. However, the logic functionality remains completely intact across all binary states, confirming robust sizing of the series PMOS structures.



# Phase 3

## Technology Node Scaling Analysis

### 3.1 3-Input Standard Cells across $0.35\mu\text{m}$ , $0.18\mu\text{m}$ , and $45\text{nm}$

#### 3.1.1 Design Objective

To verify the logic and analyze the propagation and contamination delays of 3-Input CMOS NAND and NOR gates across multiple technology nodes to observe scaling effects.

#### 3.1.2 SPICE Netlist Implementation

Below is the structural definition used for the simulations (example highlighting the  $45\text{nm}$  NOR structure).

Listing 3.1: 3-Input NOR Netlist Structure ( $45\text{nm}$ )

```

1 * PMOS network (Series connection for NOR)
2 M1 Out InA N1 Vdd PMOS l=45n w=270n
3 M2 N1 InB N2 Vdd PMOS l=45n w=270n
4 M3 N2 InC Vdd Vdd PMOS l=45n w=270n
5
6 * NMOS network (Parallel connection for NOR)
7 M4 Out InA 0 0 NMOS l=45n w=45n
8 M5 Out InB 0 0 NMOS l=45n w=45n
9 M6 Out InC 0 0 NMOS l=45n w=45n

```

#### 3.1.3 Performance Metrics

Table 3.1: Comparison of Delay Metrics (3-Input NAND)

| Tech Node         | $t_{pHL}$ (ps) | $t_{pLH}$ (ps) | Prop. Delay (ps) | Contam. Delay (ps) | Notes                     |
|-------------------|----------------|----------------|------------------|--------------------|---------------------------|
| $0.35\mu\text{m}$ | 61.7           | 31.4           | 21.6             | 20.0               | Slower, Stable            |
| $0.18\mu\text{m}$ | 37.2           | 8.6            | 22.9             | -7.6               | Best Compromise           |
| $45\text{nm}$     | 36.9           | -1.8*          | 17.5             | 19.6               | Fast, Parasitic dominance |

Table 3.2: Comparison of Delay Metrics (3-Input NOR)

| Tech Node         | $t_{pHL}$ (ps) | $t_{pLH}$ (ps) | Prop. Delay (ps) | Contam. Delay (ps) |
|-------------------|----------------|----------------|------------------|--------------------|
| $0.35\mu\text{m}$ | 33.7           | 100.6          | 67.2             | 24.0               |
| $0.18\mu\text{m}$ | -9.3*          | 50.3           | 20.5             | -32.8              |
| $45\text{nm}$     | 12.9           | 18.9           | 3.0              | 8.0                |

*\*Negative values indicate threshold-crossing artifacts in simulation due to steep submicron transitions.*

**Conclusion on Scaling:** Scaling from  $0.35\mu\text{m}$  to  $45\text{nm}$  massively improves switching speed (NOR propagation delay drops from 67 ps to 3 ps). However, advanced nodes introduce simulation artifacts and high sensitivity to parasitic capacitances. The  $0.18\mu\text{m}$  node generally offered the best balance between speed and simulation reliability in this test environment.

# Phase 4

## Pipeline Delay & Logical Effort

### 4.1 NAND2 vs. NOR2 Loading Analysis

#### 4.1.1 Design Objective

To investigate the delay of a logic pipeline consisting of a unit CMOS inverter driving a unit NAND2 (or NOR2), which in turn drives a load inverter scaled by  $\alpha = 64$ . The NOR2 gate must be scaled using Logical Effort to present equal input capacitance to the driver.

#### 4.1.2 Logical Effort Calculation

To equate the input capacitances ( $C_{\text{NAND2}} = C_{\text{NOR2}}$ ):

$$C_{\text{NAND2}} \approx 4C \quad \text{and} \quad C_{\text{NOR2}} = 5 \cdot \alpha_{\text{NOR}} \cdot C$$

$$4C = 5 \cdot \alpha_{\text{NOR}} \cdot C \quad \Rightarrow \quad \alpha_{\text{NOR}} = 0.8$$

#### 4.1.3 Pipeline Simulation Results

Table 4.1: Propagation Delay Comparison (Load  $\alpha = 64$ )

| Gate Type                        | tphl (s)               | tphl (s)               | Avg. Delay (s)         |
|----------------------------------|------------------------|------------------------|------------------------|
| NAND2 (unit)                     | $7.90 \times 10^{-10}$ | $5.41 \times 10^{-10}$ | $6.65 \times 10^{-10}$ |
| NOR2 (scaled, $\alpha = 0.8$ )   | $7.00 \times 10^{-10}$ | $5.77 \times 10^{-10}$ | $6.38 \times 10^{-10}$ |
| NOR2 (unscaled, $\alpha = 1.0$ ) | $3.81 \times 10^{-9}$  | $1.66 \times 10^{-10}$ | $1.99 \times 10^{-9}$  |

**Conclusion:** Upsizing a gate improves its ability to drive the next stage, but increases input capacitance, slowing down the previous stage. The logically scaled NOR2 ( $\alpha = 0.8$ ) balanced these effects, successfully matching the NAND2 baseline, whereas the unscaled NOR2 suffered a 3x delay penalty.

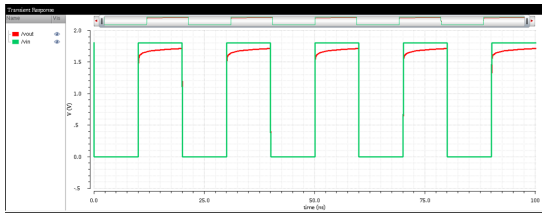
# Phase 5

## Advanced Logic Styles & Limitations

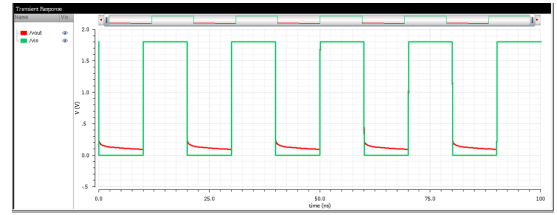
### 5.1 Pass Transistor Logic (PTL) Evaluation

#### 5.1.1 Signal Degradation Analysis

PTL uses transistors as switches to pass logic levels directly. Simulations were run to verify the threshold drops inherent in singular MOS devices.



(a) NMOS passing a Logic '1'



(b) PMOS passing a Logic '0'

Figure 5.1: Transient Response Overlaps illustrating Voltage Drops

**Observations:** \* \*\*NMOS:\*\* Successfully pulled down to 0V (Strong '0'). However, when passing  $V_{DD}$ , it saturated at  $V_{in} - V_{th}$  (Weak '1'). \* \*\*PMOS:\*\* Successfully pulled up to  $V_{DD}$  (Strong '1'). However, when passing 0V, it settled at  $|V_{th}|$  above ground (Weak '0'). \* \*\*Conclusion:\*\* Validates the absolute necessity of combining both devices into a CMOS Transmission Gate (TG) to pass full-swing logic levels without degradation.

# Phase 6

## Memory Architecture Implementation

### 6.1 4x4 NOR ROM Array Physical Design

#### 6.1.1 Architecture

In a NOR ROM, each bit is a single NMOS pull-down device. Logic '1' is the default state (bit line precharged high). The *presence* of a transistor creates a pull-down path (Logic '0'), while its *absence* leaves the bit line high (Logic '1').

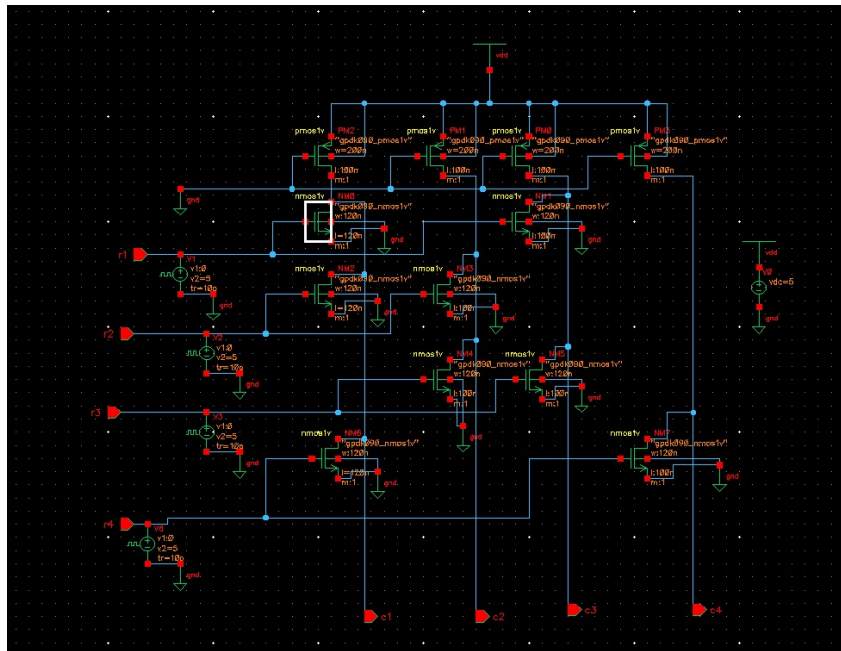


Figure 6.1: Schematic design of NOR-ROM 4x4 array

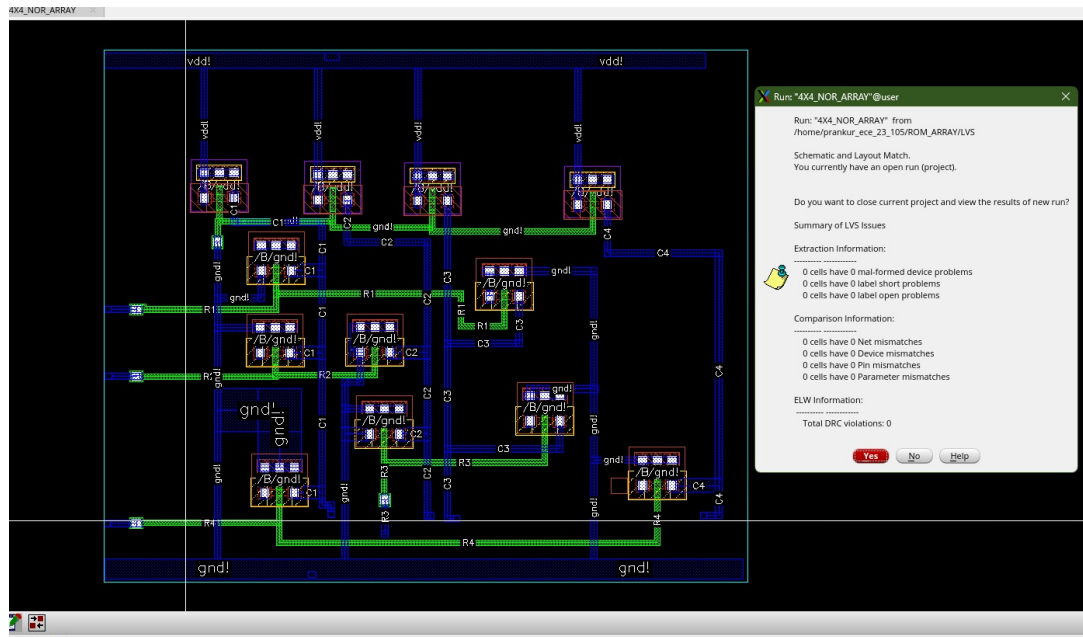


Figure 6.2: Full Custom Physical Layout of 4x4 NOR ROM

**Post-Extraction Verification:** The extracted layout showed no charge-sharing artifacts or coupling-induced disturbances. Bit-line transitions occurred strictly within word-line assertion windows, proving robust decoding isolation.

## 6.2 4x4 NAND ROM Array Verification

### 6.2.1 Architecture

A NAND ROM is the dual architecture, utilizing a series chain of MOS transistors. Logic '0' is stored by the *presence of all* transistors in the path (completing the discharge to ground). Logic '1' is stored by the *absence* of a transistor, which breaks the chain.

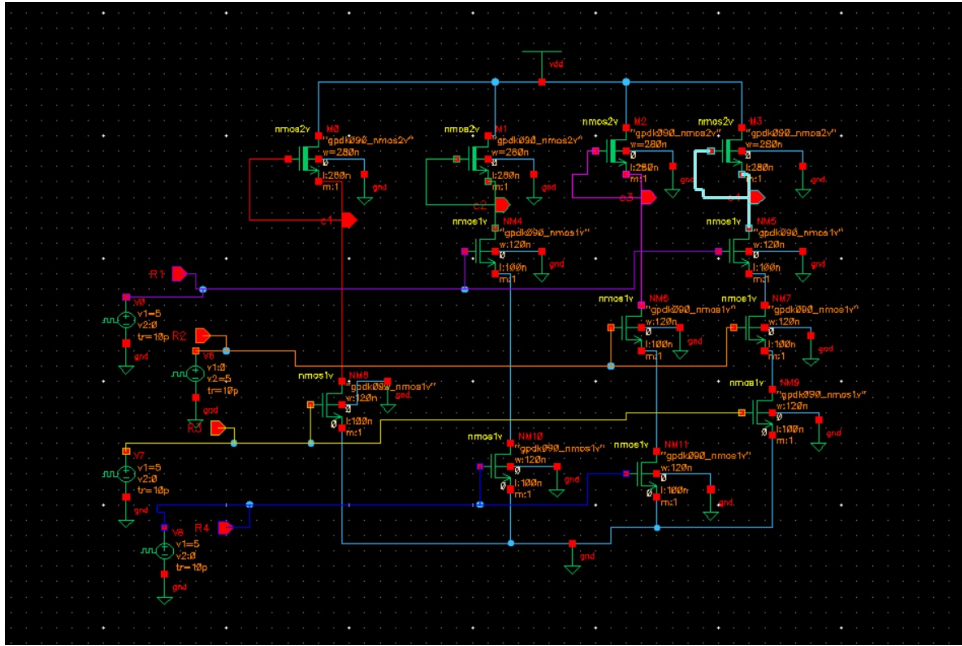


Figure 6.3: Schematic of NAND-ROM 4x4 array

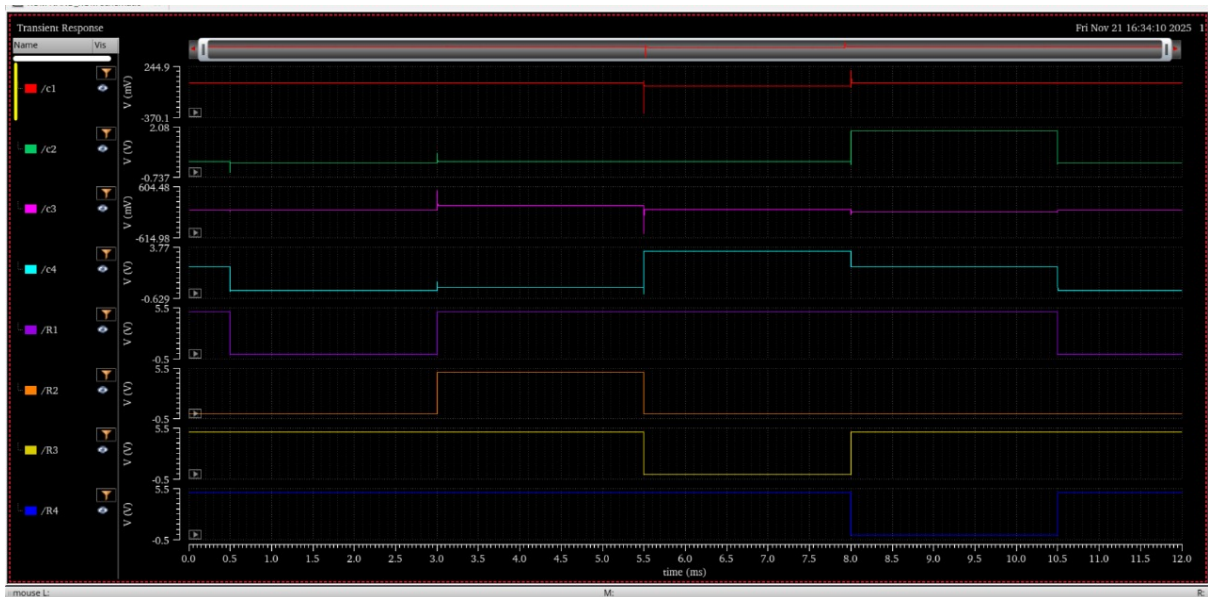


Figure 6.4: Transient Response of Word-Line Decoding

**Observations:** Sequential activation of C1–C4 demonstrated proper word-line decoding. Bit-lines discharged exclusively when a complete series path was established, validating the high-density NAND-based storage mechanism.